

Step-by-Step Guide: Attaching, Partitioning, Formatting, and Mounting an NVMe EBS Volume on AWS EC2

Overview

This document describes the **correct and recommended procedure** to create, attach, partition, format, and mount an **Amazon EBS volume** on a **Nitro-based EC2 instance**, where disks are exposed as **NVMe devices** (e.g., `/dev/nvme1n1`).

 Important: On modern EC2 instances, device names such as `/dev/xvdb` are **not used by the OS**. Always work with `/dev/nvme*` devices identified using `lsblk`.

1. Create a New EBS Volume

1. Open the AWS Management Console
 2. Navigate to **EC2 → Volumes**
 3. Click **Create volume**
 4. Select:
 5. Volume type: `gp3` / `gp2` / `io2` (as required)
 6. Availability Zone: **same AZ as the EC2 instance**
 7. Click **Create volume**
-

2. Attach the EBS Volume to the EC2 Instance

1. Select the newly created volume
2. Click **Actions → Attach volume**
3. Choose the EC2 instance
4. Device name may be shown as `/dev/xvdb` (this is only logical mapping)
5. Click **Attach**

Note: The actual device name inside the OS will be NVMe-based.

3. Identify the NVMe Device Name

Run the following command on the EC2 instance:

```
lsblk
```

Example output:

```
NAME        SIZE TYPE MOUNTPOINT
nvme0n1     30G  disk
└─nvme0n1p1 29G  part /
nvme1n1     10G  disk
```

🔗 The newly attached EBS volume is:

```
/dev/nvme1n1
```

4. Create a Partition Using `fdisk`

Run `fdisk` on the NVMe disk:

```
sudo fdisk /dev/nvme1n1
```

Inside the `fdisk` prompt:

View existing partitions

```
p
```

Create a new partition

```
n
```

- Select partition type: `p` (primary) - Partition number: `1` - First sector: **Press Enter** - Last sector: **Press Enter** (use full disk)

Write changes

```
w
```

5. Reload the Partition Table

```
sudo partprobe /dev/nvme1n1
```

Verify:

```
lsblk
```

Expected output:

```
nvme1n1
└─nvme1n1p1
```

6. Create the EXT4 File System

Format the newly created partition:

```
sudo mkfs.ext4 /dev/nvme1n1p1
```

⚠ Warning: This will erase all data on the partition.

7. Create a Mount Point

```
sudo mkdir -p /mnt/data
```

8. Mount the Partition

```
sudo mount /dev/nvme1n1p1 /mnt/data
```

Verify:

```
df -h
```

9. Configure Persistent Mount Using `/etc/fstab`

Retrieve the UUID

```
sudo blkid /dev/nvme1n1p1
```

Example:

```
UUID="a1b2c3d4-e5f6-7890" TYPE="ext4"
```

Edit `fstab`

```
sudo nano /etc/fstab
```

Add the following line:

```
UUID=a1b2c3d4-e5f6-7890 /mnt/data ext4 defaults,nofail 0 2
```

Save and exit.

10. Validate `fstab` Configuration

```
sudo mount -a
```

If no errors are shown, the configuration is valid.

11. Final Verification

```
lsblk  
df -h
```

Expected result: - `/dev/nvme1n1p1` mounted on `/mnt/data`

Key Notes & Best Practices

- Always use **UUIDs** in `/etc/fstab`
- NVMe device numbering may change after reboot
- Never assume `/dev/xvdb*` exists on Nitro instances

NVMe vs Legacy Naming

Legacy (Xen)	NVMe (Nitro)
<code>/dev/xvdb</code>	<code>/dev/nvme1n1</code>
<code>/dev/xvdb1</code>	<code>/dev/nvme1n1p1</code>

Conclusion

Following this procedure ensures a **safe, persistent, and production-ready** EBS volume setup on modern AWS EC2 instances using NVMe storage.